

Ideation Phase Brainstorm & Idea Prioritization Template

Date	15 July 2026
Team ID	XXXX
Project Name	Emotion Detection and Learning Support Engine (AI Learning Assistant)
Maximum Marks	4 Marks

Brainstorm & Idea Prioritization

As an individual contributor, this brainstorming exercise was carried out solo — problem framing, idea generation, and prioritization were all done independently, using the same structured approach a team would follow, adapted for single-author work.

Step 1: Select the Problem Statement

How might we help students who are struggling emotionally with their studies get timely, personalized support — without requiring a human tutor to be present?

Students frequently disengage, get frustrated, or lose confidence while studying, but most e-learning tools have no way to detect this and respond appropriately. This gap was the focus of the brainstorm.

Key rules followed during brainstorming: - Stay on topic - Encourage wild ideas (even if later discarded) - Defer judgment during idea generation - Go for volume before narrowing down - Be visual where possible (sketches/diagrams of flow)

Step 2: Brainstorm — Idea Listing

Ideas generated and considered during this phase:

1. A chatbot that answers subject questions directly (no emotion awareness)
2. A mood-tracking journal app for students (no AI response generation)
3. **An emotion-detection system that classifies text into learning-specific emotional states and generates personalized AI responses** ← selected
4. A voice-based stress detector during video lectures
5. A gamified reward system to boost engagement (no emotion detection)
6. A teacher-facing dashboard only, with no student-facing interaction
7. A browser extension that nudges students during long study sessions
8. A combined dual-model (BiLSTM + BERT) emotion classifier with keyword-based confidence boosting, connected to Gemini for natural language responses ← refined version of idea 3, ultimately built

Idea Grouping

- **Detection-focused ideas** (1, 3, 4, 8) — grouped as "emotion/state detection" cluster
- **Engagement-focused ideas** (2, 5, 7) — grouped as "student engagement tools" cluster
- **Dashboard-only ideas** (6) — grouped as "analytics-only" cluster

Step 3: Idea Prioritization

Ideas were placed on an Importance vs. Feasibility grid:

Idea	Importance	Feasibility	Decision
Emotion-detection + AI response system (idea 8)	High	Medium-High	o Selected – highest importance, feasible with available tools (Kaggle GPU, Gemini API, public datasets)
Mood-tracking journal (idea 2)	Medium	High	Rejected – low impact without AI-generated response
Voice-based stress detector (idea 4)	High	Low	Rejected – required audio/multimodal data and models beyond current scope/time
Gamified rewards (idea 5)	Low	High	Rejected – doesn't address the core emotional-support problem
Teacher dashboard only (idea 6)	Medium	High	Rejected – misses the direct student-support component that was the core goal

Final selection: The dual-model (BiLSTM + BERT) emotion detection system with Gemini-powered personalized responses was chosen as the highest-importance, feasible option – balancing technical achievability (public datasets, free-tier APIs, Kaggle GPU access) against genuine impact (direct, real-time emotional support for students).

Reference template: <https://www.mural.co/templates/brainstorm-and-idea-prioritization>